This article is to make readers aware about the top ten open source DevOps projects that are extensively used today. It is aimed at developers and DevOps engineers who use open source tools, so that they can improve their current software development life cycles.

As we know, DevOps is the new buzzword in the field of IT. It is nothing more than a software development practice that emphasises the collaboration and communication between software developers and IT professionals. It helps to cut down the development time and increases deployment frequency without compromising on quality. Let's take a closer look at the top ten open source tools for DevOps.

1) Git

Git is an open source, distributed version control system for tracking changes in source code during software development. Git was created for the development of the Linux kernel by Linus Torvalds in 2005, with other developers contributing to it initially. Git works on the client-server model. Each Git directory on each computer is a complete repository with full history and version tracking capabilities, independent of network access. In order to make source code accessible across the team, repositories need to be hosted on a hosting service, on which team members can push their work and go back to earlier versions if necessary. GitHub and Bitbucket are currently the two best online Git repo hosting services.

2) Jenkins

Jenkins is an open source automation server that can be used to automate all kinds of tasks related to software building, testing and deployment. Jenkins can be installed through native system packages, Docker or even run standalone on any machine with a Java Runtime Environment (JRE) installed.

Features:

- Jenkins can be used as a continuous integration server or turned into a continuous delivery hub for any project as an extensible automation server.
- It is easy to install and configure it.
- Jenkins provides hundreds of plugins to support building, deploying and automating any project. You can also develop a Jenkins plugin based on your project needs.
- Jenkins supports the master-slave architecture, i.e., it can easily distribute work across multiple machines that help you to build, test and deploy applications faster.

3) ELK

The ELK stack (Elasticsearch, Logstash and Kibana) is the most popular open source log analytics solution. The stack helps you to collect logs from different services, applications, networks and servers, and store them in a centralised location for processing and analysis. Once logs are stored in the Elasticsearch index, you can use it for analytical purposes like monitoring services, application troubleshooting, security and audit purposes, business intelligence, application performance monitoring, etc. Let's understand the role of the Elasticsearch, Logstash and Kibana stacks.

Elasticsearch: This is a highly scalable, open source, full-text search and analytics engine. It provides a distributed, multi-tenant-capable full-text search engine with a RESTful Web interface and schema-free JSON documents. Elasticsearch has been developed in Java, and is released as open source under the terms of the Apache licence.

Logstash: This is for centralised logging, log enrichment and parsing. By using Logstash, you can parse the field you're interested in from the logs, and send it to the Elasticsearch index to save the records.

Kibana: Kibana is an open source data visualisation and analytics platform designed to work with Elasticsearch. You can use it to search and view the data stored in Elasticsearch indices. You can easily perform data analysis and create interactive